MILITARY STANDARD

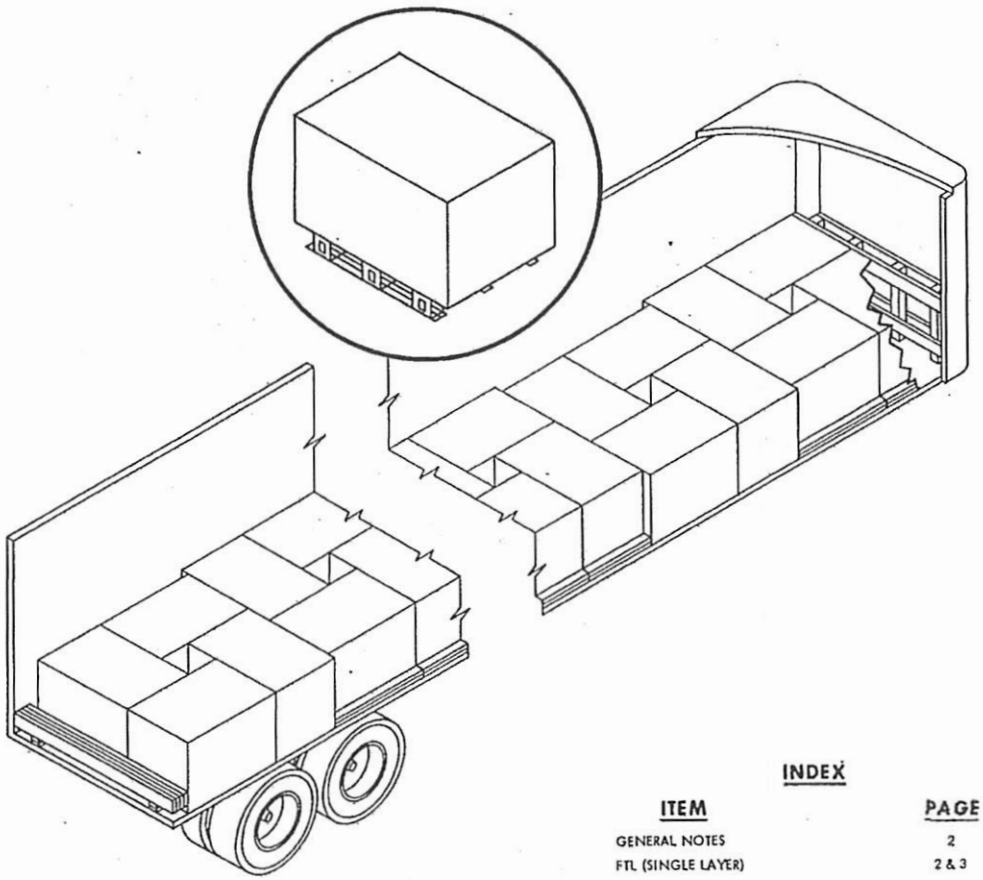
TRUCKLOADING OF HAZARDOUS MATERIALS

PALLETIZED UNIT LOADS
CHIMNEY PATTERN

MIL-STD-1320-3
(NAVY)

29 JANUARY 1975

SUPERSEDING
WR-51/3A
30 JULY 1964



INDEX

ITEM	PAGE
GENERAL NOTES	2
FTL (SINGLE LAYER)	2 & 3
FTL (MULTI-LAYER)	4 & 5
LTL	6
DETAILS	7
UNITIZING DETAIL	8

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236.

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M.S. Gray NWHL 1-29-75
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 1-29-75
SIGNATURE SEASYSCOM BY DIRECTION DATE

ORIGINATOR D.M. Osborn
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NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

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GENERAL NOTES

1. THE INDEX TO STANDARDS FOR PALLETIZING, TRUCKLOADING, RAILCAR LOADING AND CONTAINER LOADING, MIL-HDBK 236 (NAVY), LISTS HAZARDOUS MATERIALS AND INDICATES THE CORRECT DOCUMENT TO BE USED IN TRUCKLOADING OF EACH ITEM INDEXED. MIL-HDBK-236 INDEXES THE UNIT LOADS THAT ARE AUTHORIZED FOR BLOCKING AND BRACING USING THIS DOCUMENT.
2. THIS DOCUMENT PROVIDES DETAILED TRUCKLOADING INSTRUCTIONS APPLICABLE TO PALLETIZED UNIT LOADS WHEN THE SUM OF THE LENGTH AND WIDTH OF A UNIT LOAD IS LESS THAN THE WIDTH OF A STANDARD TRAILER.
3. THE PROCEDURES AND PRACTICES CONTAINED HEREIN ARE INTENDED FOR VAN TYPE TRAILERS OF ALL LENGTHS. THE TRAILERS MAY HAVE WOOD FLOORS, METAL FLOORS OR METAL FLOORS WITH WOOD NAILING STRIPS.
4. THE BLOCKING METHODS SHOWN ON PAGES 3, 5 AND 6 ARE FOR TRAILERS WITH WOOD FLOORS AND METAL FLOORS WITH WOOD NAILING STRIPS. WHEN THE TRAILER HAS METAL FLOORS OR THE CHARACTERSITICS OF THE UNIT LOAD MAKES THE BLOCKING AND BRACING SHOWN IMPRACTICABLE TO USE, CHOOSE AN APPROPRIATE ALTERNATE METHOD EXPLAINED IN THE SPECIFIC INSTRUCTIONS FOR THIS LOAD.
5. A FULL TRUCKLOAD (FTL) CONSISTS OF AS MANY UNIT LOADS THAT CAN BE ARRANGED IN THE TRAILER CHIMNEY PATTERN; CUBE, PERMISSIBLE GROSS VEHICLE WEIGHT AND AXLE LOAD LIMITATIONS PERMITTING. IF THESE LIMITATIONS PERMIT, UNIT LOADS MAY BE DOUBLE OR TRIPPLE LAYER IN ACCORDANCE WITH THE PRINCIPLE SET FORTH IN THIS DOCUMENT.
6. A LESS-THAN-TRUCKLOAD (LTL) SHOULD BE ARRANGED CONSIDERING WEIGHT DISTRIBUTION IN THE TRAILER. A TYPICAL LTL IS SHOWN ON PAGE 6.
7. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENT TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS, AND ATTACH APPROPRIATE PLACARD (IF REQUIRED) TO BOTH SIDES, FRONT AND BACK OF TRAILER.
8. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS FF-N-103, TYPE II, STYLE 10 COMMON BRIGHT; STRAPPING QQ-S-781, TYPE I, CLASS A.
9. FOR GENERAL TRUCKLOADING PROCEDURES, REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

PROCEDURE (SINGLE LAYER)

1. THE LOAD SHOWN ON PAGE 3 IS INTENDED TO ILLUSTRATE TYPICAL BLOCKING AND BRACING PROCEDURES FOR A SINGLE LAYER LOAD IN TRAILERS WITH WOOD FLOORS AND METAL FLOOR WITH WOOD NAILING STRIPS. FOR TRAILER WITH METAL FLOORS, USE THE APPROPRIATE ALTERNATE BLOCKING METHOD (SEE NOTE 3 BELOW).

WARNING

DO NOT NAIL BLOCKING OR BRACING INTO METAL FLOORS.

2. WHEN THE CHARACTERISTICS OF THE UNIT LOAD PREVENTS NAILING SLEEPERS, PIECE 6, TO THE TRAILER FLOOR, USE A FILLER ASSEMBLY, DETAIL H OR J PAGE 7, AS APPROPRIATE, ON THE SIDE OF THE UNIT LOADS.

NOTE

WHEN SLACK (TRAILER WIDTH LESS LADING WIDTH) IS LESS THAN 3 1/2 INCHES, SLEEPERS AND OR FILLER ASSEMBLIES MAY BE OMITTED.

3. WHEN THE TRAILER HAS METAL FLOORS OR METAL FLOORS WITH NAILING STRIPS AND THE STRIPS ARE NOT IN THE DESIRED LOACTION, USE A FILLER ASSEMBLY, DETAIL H OR J PAGE 7, AS APPROPRIATE, AT THE SIDE OF THE UNIT LOADS.
4. FILLER ASSEMBLIES SHALL FILL THE VOID BETWEEN THE UNIT LOADS AND TRAILER SIDE WALL. THE QUANTITY AND THICKNESS OF THE MATERIAL USED TO FABRICATE THE FILLER ASSEMBLIES MAY BE VARIED AS NECESSARY TO FILL THE VOID.
5. THE TYPE OF REAR BLOCKING TO BE USED IS DEPENDENT UPON THE DISTANCE BETWEEN THE TRAILER DOORS, WHEN CLOSED, AND THE REAR OF THE LADING.

DISTANCE LESS THAN 12 INCHES

INSTALL REAR BLOCKING ASSEMBLY, DETAIL B, AS SHOWN ON PAGE 3.

CAUTION

REAR BLOCKING MUST BEAR AGAINST LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED. DO NOT USE TRAILERS WITH ROLL UP DOORS.

DISTANCE 12 TO 36 INCHES

INSTALL REAR BLOCKING ASSEMBLY, DETAIL F, PAGE 5, AS SHOWN ON PAGE 5.

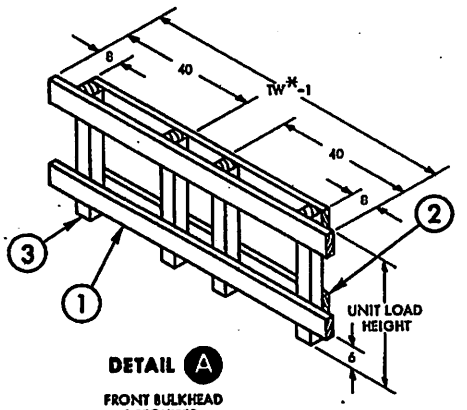
CAUTION

REAR BLOCKING MUST BEAR AGAINST LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED. DO NOT USE TRAILERS WITH ROLL UP DOORS.

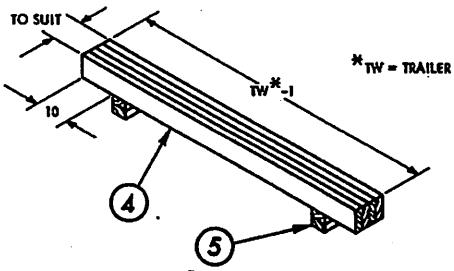
DISTANCE GREATER THAN 36 INCHES

INSTALL REAR BLOCKING AS SHOWN IN TYPICAL LTL, PAGE 6. DO NOT USE TRAILERS WITH ALL METAL FLOORS.

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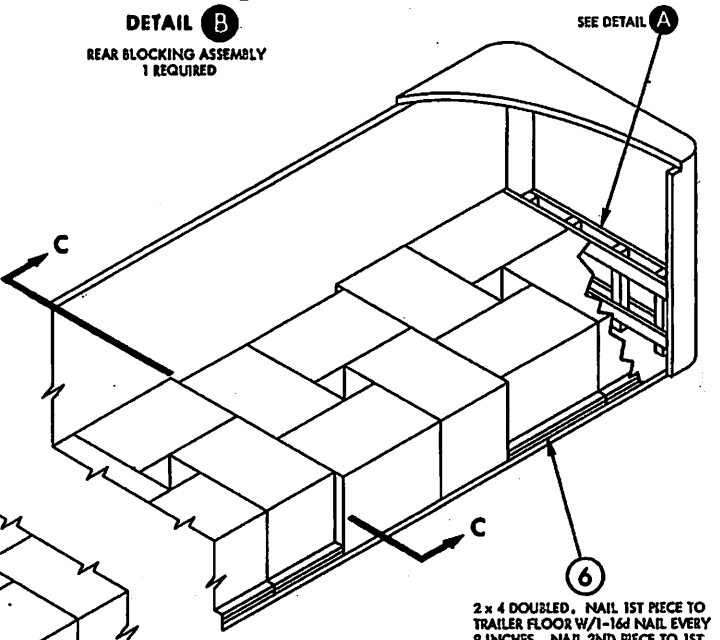
DETAIL A
FRONT BULKHEAD
1 REQUIRED



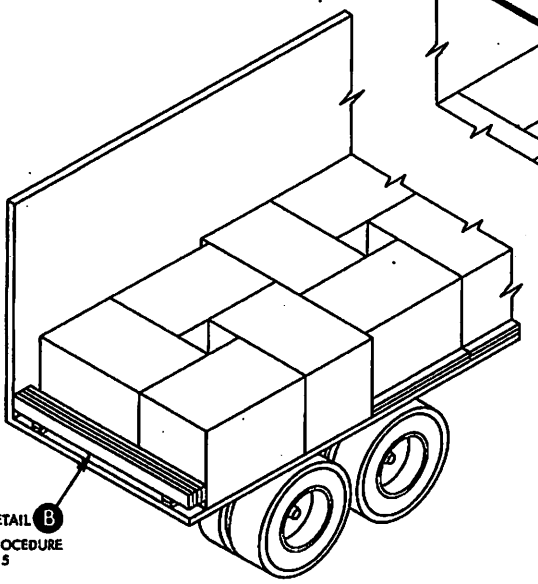
*TW = TRAILER WIDTH

DETAIL B
REAR BLOCKING ASSEMBLY
1 REQUIRED

NOTE
CHIMNEY GROUPS OF FOUR PALLETS EACH ARE LOADED AGAINST ALTERNATE SIDES OF THE TRAILER.

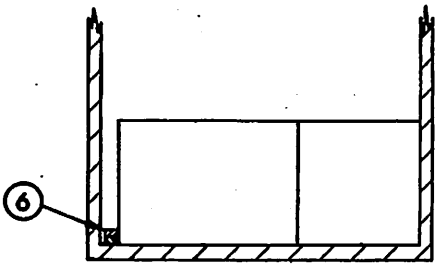


2 x 4 DOUBLED, NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.



SEE PROCEDURE NOTE 5

NOTE
THIS DRAWING ILLUSTRATES A LOAD WHEN THE DISTANCE BETWEEN THE LADING AND THE TRAILER DOORS IS 12 INCHES OR LESS. SEE PROCEDURE NOTE 5 FOR DETAILED INSTRUCTIONS FOR THIS AND OTHER CONFIGURATIONS.



SECTION C-C

*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO.PCS REQD	NAIL TO	FIELD NAILS NUMBER	FIELD NAILS SIZE	NOTE
6	SLEEPERS	2 x 4 TO SUIT	AS REQD	SEE 4	3	10d	
5	SUPPORT	2 x 4 TO SUIT	4	SEE 5	3	10d	
4	SOLID FILL	THICKNESS TO SUIT x 6 x TW*-1	AS REQD	LAMINATE	6/PIECE		TO SUIT
3	VERTICAL	4 x 4 x UNIT LOAD HT	4	SEE 1 & 2	-		-
2	FWD HORIZONTAL	2 x 6 TO SUIT	2	3	3/JOINT		12d
1	AFT HORIZONTAL	2 x 6 x TW*-1	2	3	3/JOINT		12d

LIST OF MATERIALS AND NAILING DATA

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PROCEDURE (MULTI-LAYER)

1. THE LOAD SHOWN ON PAGE 5 IS INTENDED TO ILLUSTRATE TYPICAL BLOCKING AND BRACING PROCEDURES FOR A TWO LAYER LOAD IN A TRAILER WITH WOOD, METAL OR METAL WITH WOOD NAILING STRIP FLOORS.

WARNING

DO NOT NAIL BLOCKING OR BRACING INTO METAL FLOORS.

2. TO PREVENT LONGITUDINAL MOVEMENT IN THE SECOND (AND THIRD) LAYER(S) WHERE THE LAYER CHANGES FROM TWO LAYERS TO ONE LAYER HIGH (OR THREE LAYERS TO TWO LAYERS HIGH) AND AT THE REAR OF THE LOAD WHEN TWO OR MORE LAYERS HIGH, THE UNIT LOADS SHALL BE UNITIZED AS SHOWN ON PAGE 8. THE STIFFENERS SHALL BE POSITIONED ON THE UNIT LOADS SO THAT WHEN THE UNIT LOADS ARE IN THE TRAILER, THE STRAPPING IS PARALLEL TO THE LONGITUDINAL AXIS OF THE TRAILER. THE STIFFENERS SHALL ALSO BE POSITIONED TOWARDS THE LOWER LAYER(S) AND WHEN AT THE REAR OF THE LOAD, TOWARDS THE REAR.
3. USE FILLER ASSEMBLIES, DETAILS H, J, K OR L, AS APPROPRIATE TO FILL THE VOIDS BETWEEN THE UNIT LOADS AND THE TRAILER SIDE WALLS. THE QUANTITY AND THICKNESS OF THE MATERIAL USED TO FABRICATE THE FILLER ASSEMBLIES MAY BE VARIED AS NECESSARY TO FILL THE VOID.

NOTE

WHEN SLACK (TRAILER WIDTH LESS LADING WIDTH) IS LESS THAN 3 1/2 INCHES, FILLER ASSEMBLIES MAY BE OMITTED.

4. WHEN TRAILER HAS FLOORS OF WOOD OR METAL WITH WOOD NAILING STRIPS, DOUBLED 2 x 4 SLEEPER MAY BE SUBSTITUTED FOR THE SPACER ASSEMBLY, DETAIL H, WHERE THE UNIT LOADS ARE ONE LAYER HIGH.
5. THE TYPE OF REAR BLOCKING TO BE USED IS DEPENDENT UPON THE DISTANCE BETWEEN THE LADING AND THE TRAILER DOORS WHEN CLOSED.

DISTANCE LESS THAN 12 INCHES

INSTALL REAR BLOCKING ASSEMBLY, DETAIL B, AS SHOWN ON PAGE 3.

CAUTION

REAR BLOCKING MUST BEAR AGAINST THE LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED. DO NOT USE TRAILERS WITH ROLL UP DOORS.

DISTANCE 12 TO 36 INCHES

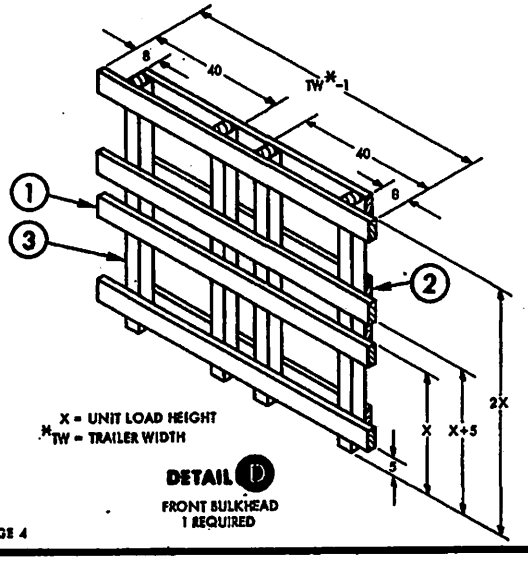
INSTALL REAR BLOCKING ASSEMBLY, DETAIL F, AS SHOWN ON PAGE 5.

CAUTION

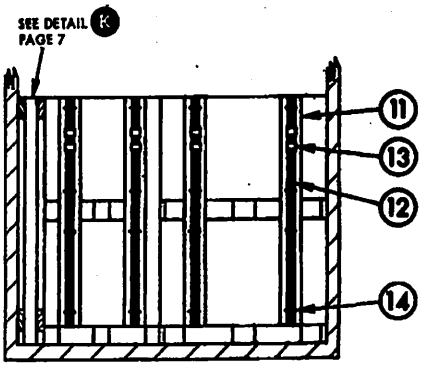
REAR BLOCKING MUST BEAR AGAINST THE LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED. DO NOT USE TRAILERS WITH ROLL UP DOORS.

DISTANCE GREATER THAN 36 INCHES

INSTALL BLOCKING AS SHOWN IN TYPICAL LTL, PAGE 6.



DETAIL D
FRONT BULKHEAD
1 REQUIRED

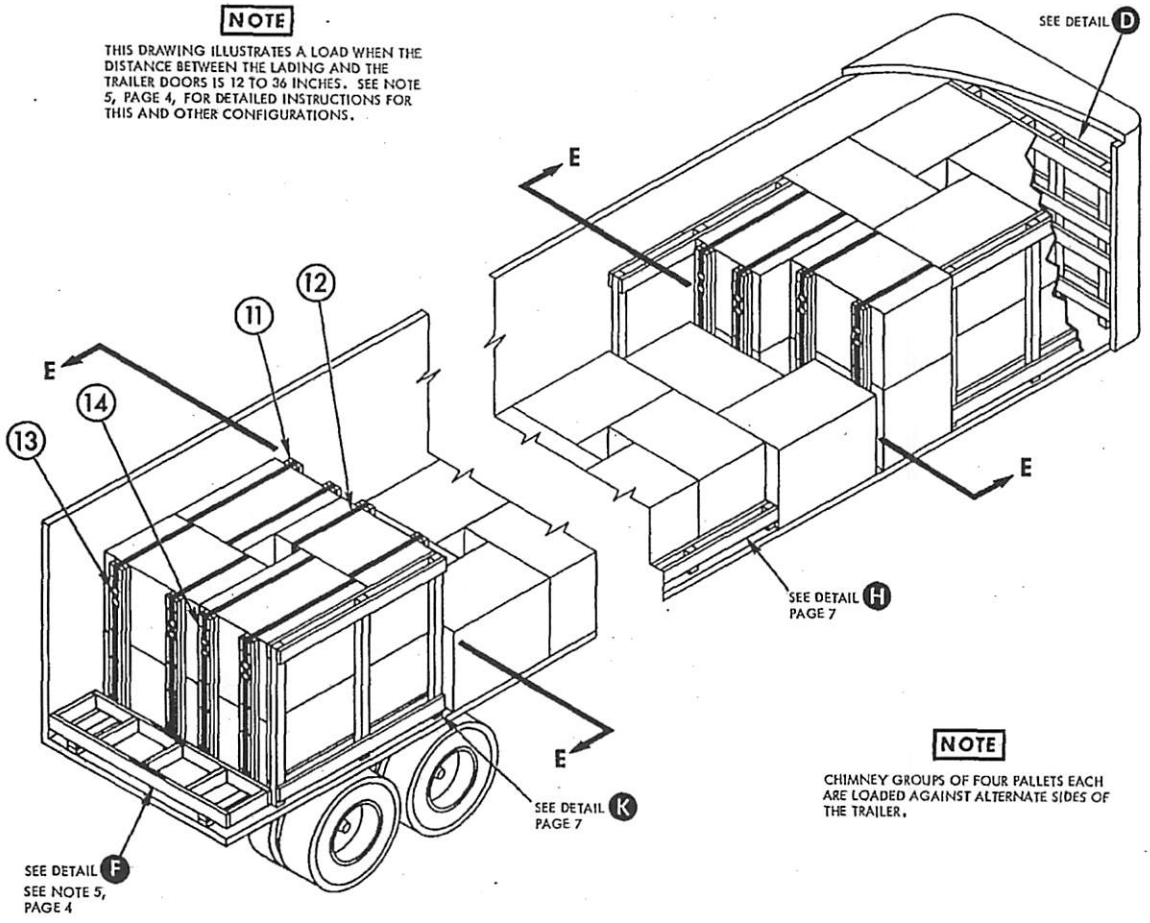


SECTION E-E

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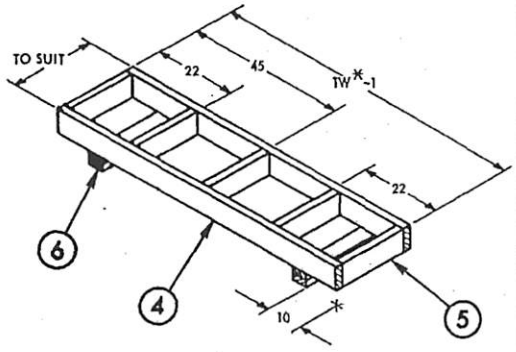
NOTE

THIS DRAWING ILLUSTRATES A LOAD WHEN THE DISTANCE BETWEEN THE LADING AND THE TRAILER DOORS IS 12 TO 36 INCHES. SEE NOTE 5, PAGE 4, FOR DETAILED INSTRUCTIONS FOR THIS AND OTHER CONFIGURATIONS.



NOTE

CHIMNEY GROUPS OF FOUR PALLETS EACH ARE LOADED AGAINST ALTERNATE SIDES OF THE TRAILER.



DETAIL F

REAR BLOCKING ASSEMBLY
1 REQUIRED

*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER NAILS	SIZE NAILS
14	STAPLE	FOR 1 1/4 STRAP	48	-	-	-
13	SEAL	FOR 1 1/4 STRAP	24	-	-	-
12	STRAP	1 1/4 x .035	12	-	-	-
11	STIFFNER	2 x 6 TO SUIT	24	DBL LAMINATE	4	8d
10	VERTICAL	4 WIDE x THICKNESS TO SUIT	AS REQD	SEE 9	-	-
9	HORIZONTAL	6 WIDE x THICKNESS TO SUIT	AS REQD	10	3/JOINT	TO SUIT
8	VERTICAL	4 WIDE x THICKNESS TO SUIT x 10	AS REQD	SEE 7	-	-
7	HORIZONTAL	6 WIDE x THICKNESS TO SUIT	AS REQD	8	3/JOINT	TO SUIT
6	SUPPORT	2 x 4 TO SUIT	4	6 5	2/JOINT 3	12d 12d
5	STRUT	2 x 6 TO SUIT	5	SEE 4	-	-
4	CROSSMEMBER	2 x 6 x TW*-1	2	5	3/JOINT	16d
3	VERTICAL	4 x 4 TO SUIT	4	SEE 1 & 2	-	-
2	FWD HORIZONTAL	2 x 6 TO SUIT	4	3	3/JOINT	12d
1	AFT HORIZONTAL	2 x 6 x TW*-1	4	3	3/JOINT	12d

LIST OF MATERIALS AND NAILING DATA

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LTL (ALL LENGTH TRAILERS)

1. THE LTL SHOWN ON THIS PAGE IS INTENDED TO ILLUSTRATE TYPICAL BLOCKING AND BRACING METHODS FOR LTL IN A TRAILER HAVING WOOD FLOORS OR METAL FLOORS WITH WOOD NAILING STRIPS.

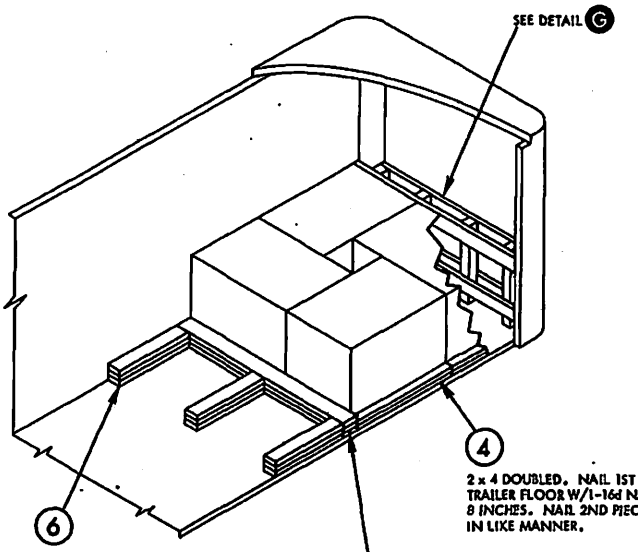
WARNING

DO NOT USE ALL METAL FLOOR TRAILERS FOR LTL SHIPMENTS AS THEY CAN NOT BE BLOCKED BY THIS METHOD.

2. FILLER ASSEMBLIES, DETAIL H AND J, MAY BE USED AS AN ALTERNATE FOR SLEEPER, PIECE 4.
3. AN LTL MAY CONSIST OF MORE OR LESS UNIT LOADS AND ARE NOT NECESSARILY LIMITED TO THE AMOUNT SHOWN.

NOTE

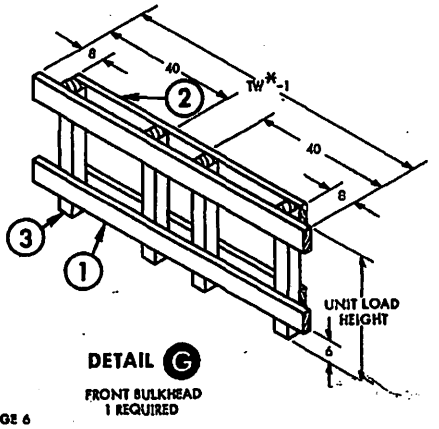
THIS DRAWING ILLUSTRATES A LOAD WHEN THE DISTANCE BETWEEN THE LADING AND THE TRAILER DOORS IS GREATER THAN 36 INCHES, THE TRAILER SHALL HAVE WOOD FLOORS OR METAL FLOORS WITH NAILING STRIPS.



2 x 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

2 x 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.

2 x 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/9-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.



DETAIL G

FRONT BULKHEAD 1 REQUIRED

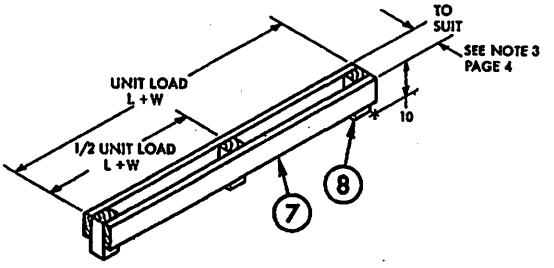
*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS	
					NUMBER	SIZE
6	BACKUP	2 x 6 x 36	9	SEE	FIELD	NOTE
5	CROSSMEMBER	2 x 6 x TW*-1	3	SEE	FIELD	NOTE
4	SLEEPER	2 x 4 TO SUIT	AS REQD	SEE	FIELD	NOTE
3	VERTICAL	4 x 4 x UNIT LOAD HT	4	SEE 1 & 2	-	-
2	FWD HORIZONTAL	2 x 6 TO SUIT	2	3	3/JOINT	12d
1	AFT HORIZONTAL	2 x 6 x TW*-1	2	3	3/JOINT	12d

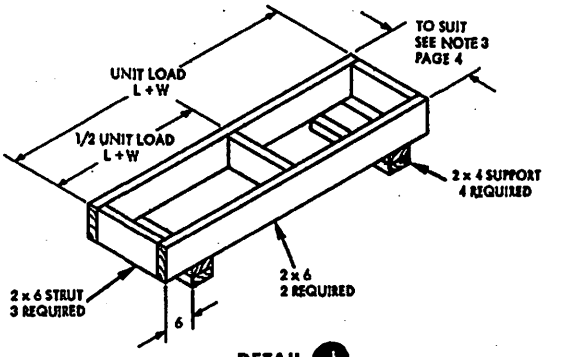
LIST OF MATERIALS AND NAILING DATA

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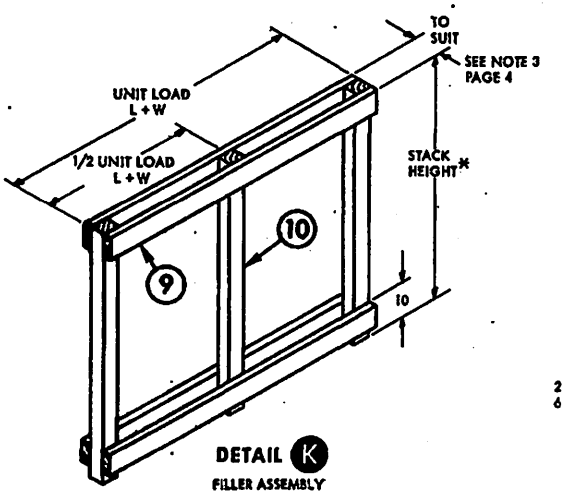
DETAILS (FILLER ASSEMBLIES)



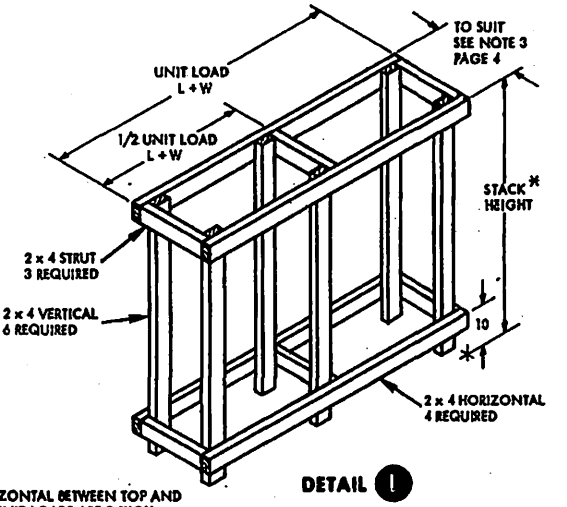
DETAIL H
FILLER ASSEMBLY



DETAIL J
FILLER ASSEMBLY



DETAIL K
FILLER ASSEMBLY



DETAIL L
FILLER ASSEMBLY

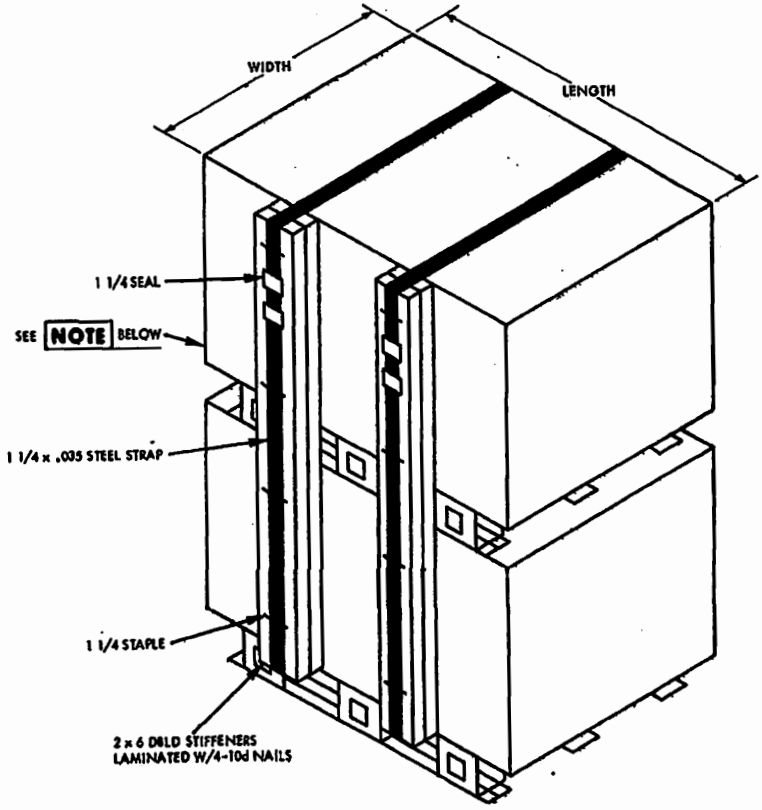
*INSTALL THIRD SET OF HORIZONTAL BETWEEN TOP AND BOTTOM HORIZONTALS IF UNIT LOADS ARE 3 HIGH.

NOTE

WHEN REQUIRED TO FILL A LARGE VOID, DETAIL J MAY BE SUBSTITUTED FOR DETAIL H AND DETAIL L MAY BE SUBSTITUTED FOR DETAIL K.

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UNITIZING DETAIL



1. WHEN REQUIRED BY THE FTL OR LTL REQUIREMENTS OF THIS MIL-STD, STACKED UNIT LOADS SHALL BE UNITIZED.

WARNING

THE CHIMNEY PATTERN REQUIRES UNITIZING IN TWO CONFIGURATIONS: WITH THE STIFFENERS ON THE "LENGTH" SIDE OF THE STACKED UNIT LOADS AS SHOWN ABOVE AND STIFFENERS ON THE "WIDTH" SIDE. UNITIZING SHALL BE ACCOMPLISHED SO THAT THE REQUIREMENTS OF NOTE 2, PAGE 4 ARE SATISFIED.

2. THE DOUBLED 2 x 6 STIFFENERS SHALL EXTEND FROM THE TOP OF THE STACKED UNIT LOADS TO THE PALLET OF THE BOTTOM UNIT LOAD.
3. THE 1 1/4 INCH STEEL STRAPS POSITIONED AS SHOWN, ENCIRCLE THE STACKED UNIT LOADS AND PASS UNDER THE TOP DECK OF THE BOTTOM PALLET.
4. TENSION STRAPS AND SEAL WITH TWO DOUBLE CRIMPED SEALS. SECURE EACH STRAP TO THE STIFFENER WITH FOUR 1 1/4 INCH STRAP STAPLES.

NOTE

WHEN LOADING A TRAILER WITH A SINGLE STACK OF UNIT LOADS, THE STIFFENERS SHALL BE AT BOTH ENDS OF THE LOAD.

5. WHEN UNITIZING THREE HIGH, FOLLOW THE PRINCIPLES OF THE TWO HIGH UNIT LOADS SHOWN ABOVE.

REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N281)

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